

Definition

A transition element is a d-block element whose atom or cation contains _____

Atomic trends

atomic radii	1 st I.E.
atomic radii of transition elements across the period _____ _____	1 st I.E. of transition elements across the period _____ _____
Explanation: across the row of transition element; ① shielding effect (no. of inner d e ⁻) ② nuclear charge ③ electrostatic attraction between nucleus and outermost e ⁻ ⇒ e ⁻ cloud size	Explanation: across the row of transition element; ① shielding effect (no. of inner d e ⁻) ② nuclear charge ③ electrostatic attraction between nucleus and <u>outermost e⁻</u> ⇒ energy req. to remove outermost e ⁻

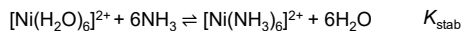
Comparison with s-block

	property	transition element	s-block element	explanation
Physical	melting point			➤ 3d and 4s e ⁻ in transition elements are available for _____ into sea of e ⁻ → stronger metallic bonding for transition elements
	density			➤ transition metals have smaller atomic size and larger atomic mass → greater mass per unit volume
	oxidation states			➤ in transition elements, the energy level of the 3d and 4s e ⁻ are _____ → both the 3d and 4s e ⁻ are available for bond formation (ionic or covalent)
Chemical	colour of cpd / ions	coloured	(s) = white (aq) = colourless	➤ due to d-d transition in transition elements (refer to explanation below)
	catalytic activity	often	little/ none	➤ <i>homogenous catalyst</i> : transition elements can exist in different oxidation states & there is ease in converting from one oxidation state to another ➤ <i>heterogenous catalyst</i> : transition elements contain partially filled 3d subshell / contain empty low-lying vacant 3d orbitals to form weak bonds with reactant molecules which is required in heterogenous catalysis
	tendency for complex formation			➤ transition elements have low-lying vacant orbitals to accommodate the lone pairs of e ⁻ from ligands to form co-ordinate bonds ➤ transition elements are relatively small and highly charged → high charge density to attract the lone pair of e ⁻ from the ligands

Complex & Ligands

Complex	A complex contains a central metal atom or ion linked to one or more surrounding ions or molecules (called ligands) by _____
Ligands	A ligand is an ion or a molecule which contains at least one atom bearing a lone pair of electrons which can be donated into a _____ of a central metal atom or ion forming a dative covalent bond. type of ligands: monodentate, bidentate, tridentate, hexadentate
co-ordination number	In a complex, the co-ordination number of the central metal atom or ion is the total number of _____ that the central metal atom or ion has formed with ligands.

Ligand Exchange



What does a large K_{stab} value suggest about the ligands?

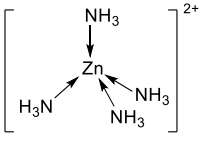
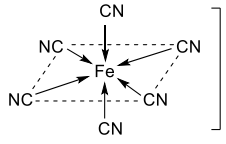
- POE lies towards the _____
- $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is relatively _____ stable than $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
- NH_3 is a _____ ligand than H_2O
- NH_3 forms a _____ dative covalent bond with Ni^{2+} compared to H_2O
- the _____ ligands are readily replaced by _____ ligands

How does the stability of Co(III) with respect to Co(II) in equation 1 compare to equation 2?

- ① $[\text{Co}(\text{H}_2\text{O})_6]^{3+} + \text{e}^- \rightleftharpoons [\text{Co}(\text{H}_2\text{O})_6]^{2+} \quad E^\ominus = +1.82 \text{ V}$
- ② $[\text{Co}(\text{NH}_3)_6]^{3+} + \text{e}^- \rightleftharpoons [\text{Co}(\text{NH}_3)_6]^{2+} \quad E^\ominus = +0.10 \text{ V}$

- Co(III) is _____ stable in eqⁿ 1
- the _____ $E^\ominus$ value in eqⁿ 1 show that Co(III) reduces _____ readily to Co(II) in eqⁿ 1

d orbital splitting & d-d transition

complex		
energy level of d orbitals		
explaining how d orbitals are split	• in a tetrahedral complex, the ligands approach the metal ion _____ the x, y and z axes • $3d_{xy}$, $3d_{yz}$, $3d_{zx}$ orbitals have greatest electron density _____ the x, y and z axes, hence repulsion between e⁻ in the orbitals and lone pairs of the approaching ligand is _____ • $3d_{x^2-y^2}$, $3d_{z^2}$ orbitals have greatest electron density _____ the x, y and z axes, hence the repulsion is _____	• in an octahedral complex, the ligands approach the metal ion _____ the x, y and z axes • $3d_{xy}$, $3d_{yz}$, $3d_{zx}$ orbitals have greatest electron density _____ the x, y and z axes, hence repulsion between e⁻ in the orbitals and lone pairs of the approaching ligand is _____ • $3d_{x^2-y^2}$, $3d_{z^2}$ orbitals have greatest electron density _____ the x, y and z axes, hence the repulsion is _____
explanation for colour	• presence of _____ causes the _____ • _____ of the five 3d orbitals in the transition metal ion into two sets of different energy levels • since the 3d subshell is _____, • _____ from the lower-energy d orbitals can absorb energy corresponding to certain wavelengths from the visible spectrum and get promoted to the higher-energy d orbitals. (d-d transitions) • the _____ observed is the compliment of the colour absorbed.	
factors affecting ΔE	① ②	

Definition

A transition element is a d-block element whose atom or cation contains incomplete d-subshell

Atomic trends

atomic radii	1 st I.E.
atomic radii of transition elements across the period <u>remains</u> <u>approximately constant</u>	1 st I.E. of transition elements across the period <u>remains</u> <u>approximately constant</u>
Explanation: across the row of transition element; ① shielding effect (no. of inner d e ⁻) ② nuclear charge ③ electrostatic attraction between nucleus and outermost e ⁻ ⇒ e ⁻ cloud size	Explanation: across the row of transition element; ① shielding effect (no. of inner d e ⁻) ② nuclear charge ③ electrostatic attraction between nucleus and <u>outermost e⁻</u> ⇒ energy req. to remove outermost e ⁻

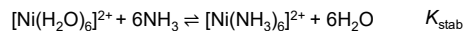
Comparison with s-block

	property	transition element	s-block element	explanation
Physical	melting point	higher	lower	➤ 3d and 4s e ⁻ in transition elements are available for <u>delocalisation</u> into sea of e ⁻ → stronger metallic bonding for transition elements
	density	higher	lower	➤ transition metals have smaller atomic size and larger atomic mass → greater mass per unit volume
	oxidation states	variable	fixed	➤ in transition elements, the energy level of the 3d and 4s e ⁻ are <u>similar</u> → both the 3d and 4s e ⁻ are available for bond formation (ionic or covalent)
Chemical	colour of cpd / ions	coloured	(s) = white (aq) = colourless	➤ due to d-d transition in transition elements (refer to explanation below)
	catalytic activity	often	little/ none	➤ <i>homogenous catalyst</i> : transition elements can exist in different oxidation states & there is ease in converting from one oxidation state to another ➤ <i>heterogenous catalyst</i> : transition elements contain partially filled 3d subshell / contain empty low-lying vacant 3d orbitals to form weak bonds with reactant molecules which is required in heterogenous catalysis
	tendency for complex formation	high	low	➤ transition elements have low-lying vacant orbitals to accommodate the lone pairs of e ⁻ from ligands to form co-ordinate bonds ➤ transition elements are relatively small and highly charged → high charge density to attract the lone pair of e ⁻ from the ligands

Complex & Ligands

Complex	A complex contains a central metal atom or ion linked to one or more surrounding ions or molecules (called ligands) by <u>co-ordinate bonds/ dative covalent bonds</u>
Ligands	A ligand is an ion or a molecule which contains at least one atom bearing a lone pair of electrons which can be donated into a <u>low-lying vacant orbital</u> of a central metal atom or ion forming a dative covalent bond. type of ligands: monodentate, bidentate, tridentate, hexadentate
co-ordination number	In a complex, the co-ordination number of the central metal atom or ion is the total number of <u>co-ordinate bonds/ dative covalent bonds</u> that the central metal atom or ion has formed with ligands.

Ligand Exchange



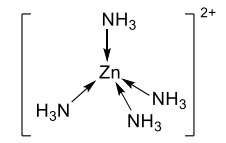
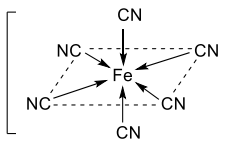
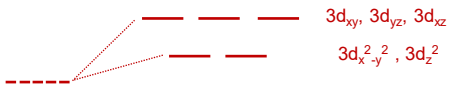
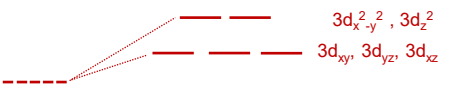
What does a large K_{stab} value suggest about the ligands?

- POE lies towards the right
- $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is relatively more stable than $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
- NH_3 is a stronger ligand than H_2O
- NH_3 forms a stronger dative covalent bond with Ni^{2+} compared to H_2O
- the H_2O ligands are readily replaced by NH_3 ligands

How does the stability of Co(III) with respect to Co(II) in equation 1 compare to equation 2?

- ① $[\text{Co}(\text{H}_2\text{O})_6]^{3+} + \text{e}^- \rightleftharpoons [\text{Co}(\text{H}_2\text{O})_6]^{2+} \quad E^\ominus = +1.82 \text{ V}$
 - ② $[\text{Co}(\text{NH}_3)_6]^{3+} + \text{e}^- \rightleftharpoons [\text{Co}(\text{NH}_3)_6]^{2+} \quad E^\ominus = +0.10 \text{ V}$
- Co(III) is less stable in eqⁿ 1
 - the more positive $E^\ominus$ value in eqⁿ 1 show that Co(III) reduces more readily to Co(II) in eqⁿ 1

d orbital splitting & d-d transition

complex	 tetrahedral	 octahedral
energy level of d orbitals	 $3d_{xy}, 3d_{yz}, 3d_{xz}$ $3d_{x^2-y^2}, 3d_{z^2}$	 $3d_{x^2-y^2}, 3d_{z^2}$ $3d_{xy}, 3d_{yz}, 3d_{xz}$
explaining how d orbitals are split	• in a tetrahedral complex, the ligands approach the metal ion <u>in-between</u> the x, y and z axes • $3d_{xy}, 3d_{yz}, 3d_{xz}$ orbitals have greatest electron density <u>in-between</u> the x, y and z axes, hence repulsion between e⁻ in the orbitals and lone pairs of the approaching ligand is <u>greater</u> • $3d_{x^2-y^2}, 3d_{z^2}$ orbitals have greatest electron density <u>in-between</u> the x, y and z axes, hence the repulsion is <u>lesser</u>	• in an octahedral complex, the ligands approach the metal ion <u>along</u> the x, y and z axes • $3d_{xy}, 3d_{yz}, 3d_{xz}$ orbitals have greatest electron density <u>in-between</u> the x, y and z axes, hence repulsion between e⁻ in the orbitals and lone pairs of the approaching ligand is <u>lesser</u> • $3d_{x^2-y^2}, 3d_{z^2}$ orbitals have greatest electron density <u>along</u> the x, y and z axes, hence the repulsion is <u>greater</u>
explanation for colour	• presence of <u>ligands</u> causes the • <u>splitting</u> of the five 3d orbitals in the transition metal ion into two sets of different energy levels • since the 3d subshell is <u>partially filled</u>, • <u>electron</u> from the lower-energy d orbitals can absorb energy corresponding to certain wavelengths from the visible spectrum and get promoted to the higher-energy d orbitals. (d-d transitions) • the <u>colour</u> observed is the compliment of the colour absorbed.	
factors affecting ΔE	① <u>identity of metal & oxidation state</u> ② <u>identity of ligand</u>	